

boiler feed tank. A high level feed tank will enable the pumps to handle warm feed water and can improve the overall efficiency of the plant. A typical pumping trap is shown in Figure 4.23.

When the trap is empty, the exhaust valve C is open and the steam valve D is closed. Water flowing into the trap through the non-return valve A raises the float E until it compresses the spring H, exerting a force which plucks the spindle J away from the magnet G so that the exhaust valve is projected into the closed position and the steam valve D is opened through the movement of the lever. Steam is therefore admitted to the trap and drives out the water through the non-return valve B. The float falls with the level of the water inside the trap and engages the collar F, pulling the spindle J down so that the exhaust valve opens and the steam valve closes. The cycle of operation is then repeated. Being float controlled, the trap operates only when water is flowing into it.

Maintenance of traps

A defective trap wastes steam and fails to ensure adequate heating. Poor heater performance can sometimes be traced to a defective steam trap. One remedy is to close in the steam outlet valve to perhaps half or quarter of a turn. Most traps can be regularly opened for inspection. The only exception is the thermodynamic type. When operating correctly this gives a characteristic 'click', usually at intervals of between 20 and 30 s so that its performance can be checked simply by listening.

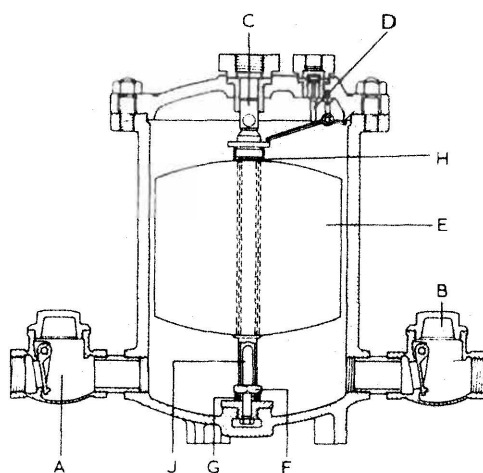


Figure 4.23 *Automatic pumping or vacuum trap (Royles Ltd.)*

- | | |
|-------------------------|------------|
| A, B. Non-return valves | F. Collar |
| C. Exhaust valve | G. Magnet |
| D. Steam valve | H. Spring |
| E. Float | J. Spindle |

Before inspecting any trap, it is advisable to check the strainer which should always be provided at the inlet of the trap. The contents of the strainer screen will give an indication of the cleanliness of the system. Fine dirt passing through the strainer is one of the chief causes of defective steam traps. Pieces of pipe scale or dirt jammed across the seat can prevent proper closure and if left for a period of time can give rise to wire drawing.

In most cases cleaning and reassembling should be sufficient to ensure satisfactory operation. If the thermodynamic type fails to operate after cleaning, then it could be that the disc and seat require lapping. Mechanical traps should be checked for defective floats and buckets and wear on any linkages, while the valves and seats should be renewed if necessary. In the case of thermostatic traps, the elements should be checked to ensure that they are sound. Particular care should be taken with the elements of bimetallic steam traps. Incorrect adjustment can give rise to excessive waterlogging or cause the traps to blow steam, while the elements themselves can sometimes assume a permanent 'set'.

Strainers and filters

The term strainer is sometimes used specifically for a simple device made up with a single layer of coarse gauze, a very coarse wire mesh or a drilled or perforated plate. The strainer in this sense is installed to hold large foreign objects which could cause damage or blockage. The term filters also describes a device designed to prevent the passage of unwanted solids into or further along a system, but the implication is that the particles are very small and the filter cartridge or element has a more complex or elaborate make up.

The simplest strainer consists of a box with a removable lid in which a flat perforated plate is inserted such that the fluid must pass through the perforations. Such strainers are found in bilge systems immediately before bilge valves. Perforation sizes vary according to duty and manufacture but are usually in the range 3–12 mm.

Strainer plates corrode and erode and when cleaning due attention should be paid to the condition of the plate. If a gap has formed at the top or bottom of the plate (so that solid objects or rags could pass through) or if it is in danger of breaking up then it should be renewed. These strainers protect the pump. If strainer covers are hinged the hinges are made loose to avoid the problem of poor seating. The state of the gasket should be checked, mating surfaces must be clean and care should be taken when closing the cover. Since these covers are frequently of cast iron and are secured by lugs and thumb screws care must be taken not to use undue force when closing them.

For high pressure water, steam or oil services simple basket strainers are used. These consist of a cylindrical container in which a perforated metal or wire basket is suspended (shown left Figure 4.24). Flow through these units is from the top, into the basket and out from the outside of the basket. They may be installed as duplex units with three way cocks at inlet and outlet so that one or both baskets can be in use, but one can be shut down for cleaning. Close to

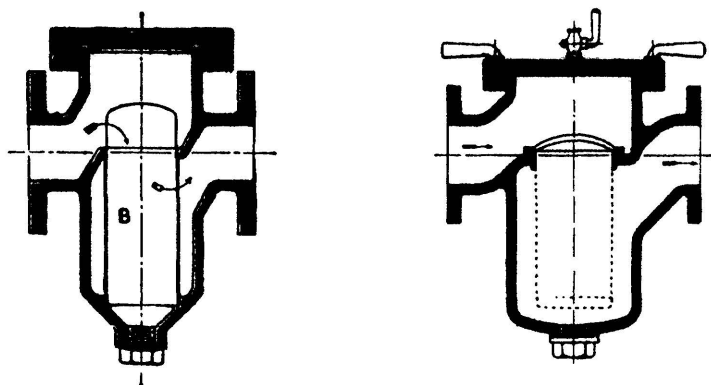


Figure 4.24 Single strainers for (left) high pressure water, steam or oil service and (right) low pressure water or oil service

sea-water suction valves, similar basket strainers (shown right in Figure 4.24) having air release cocks, are installed.

Lubricating oil systems are fitted with a wide variety of strainers some of which can be cleaned *in situ*. The knife edge strainer, (Figure 4.25) has a series of discs ganged to a shaft. Interspaced between the discs are a number of thin fingers. The solid particles are trapped on the edges of, and between, the discs. By rotating the disc shaft the particles are cleared by the fingers and fall to a sump, which is drained periodically. It is essential to operate these strainers regularly to prevent clogging when rotation may be difficult. Some have been fitted with electric motors for continuous operation.

Cartridge filters (Figure 4.26) are usually of the duplex type and have various types of element. Some cartridges can be removed for cleaning; others

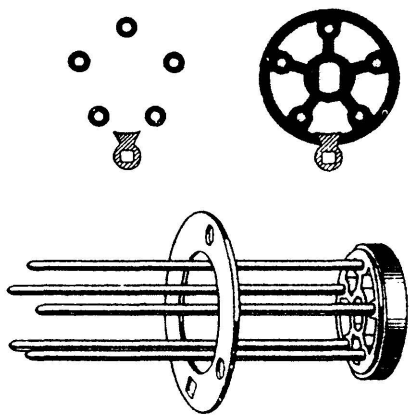


Figure 4.25 The Auto-Klean strainer showing circular straining plates, separation washers and spider (Auto-Klean Strainers Ltd.)

Wire mesh cartridge
filter

Air cock for venting
not shown

Detail of
cartridge

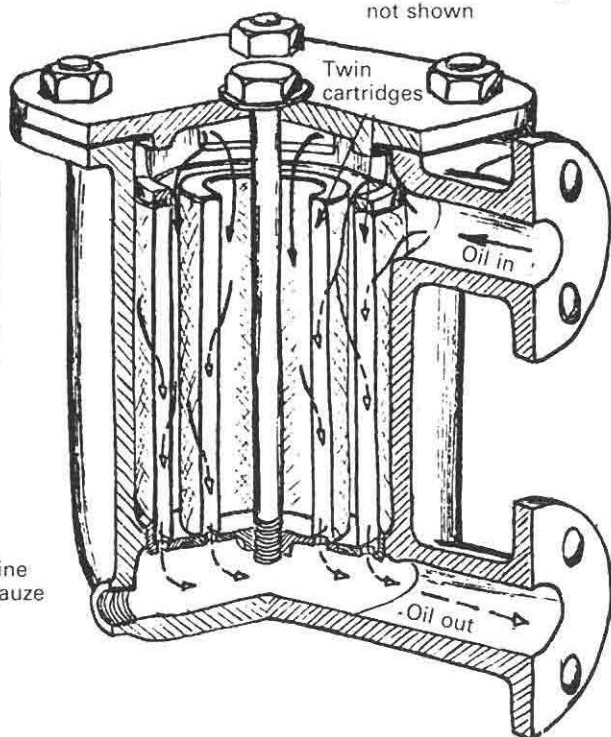
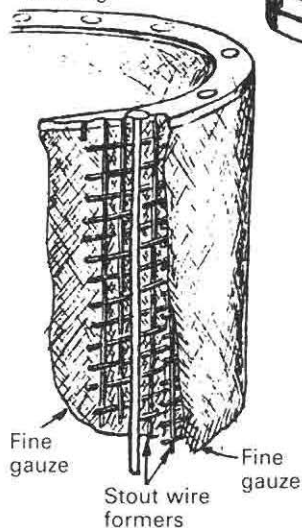


Figure 4.26 *Cartridge filters*

have renewable elements. Cartridges can be of gauze layers with different mesh sizes others are made up with layers of mesh and felt, some use membrane type materials.

Magnetic filters (Figure 4.27) provide extra protection for engines and gearboxes where iron or steel wear particles are likely to be present.

Further reading

Milton, J. H. and Leach, R. M. (1980) *Marine Steam Boilers*, 4th edn. Butterworths.

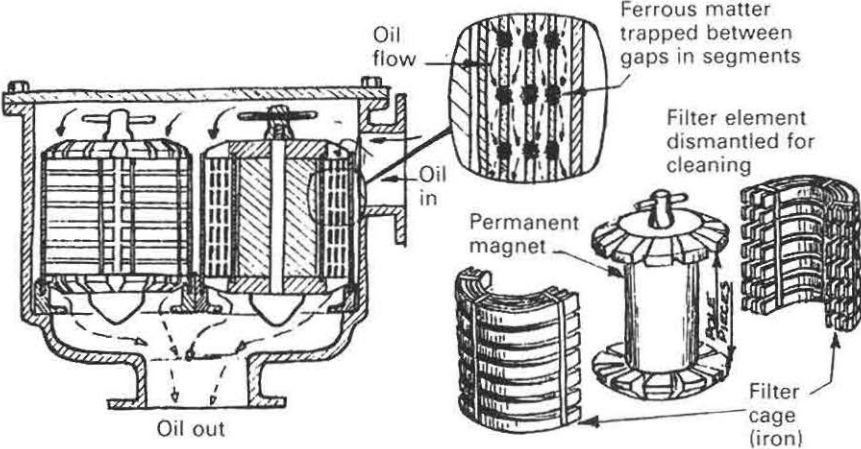


Figure 4.27 *Magnetic filter*